

Task 2:

1. Agentic Ai:

An agentic ai could be defined as group of agents or multiple systems collaborating together with each agent having a unique purpose and work on common goal.

Example : in an agentic system , 1 agent is responsible for surfing the internet and gathering / scrapping information . this information would be then utilized by another agent (agent 2) to summarize and make the data or information more understandable . 3rd agent (with tool) may then send this information to other(lets say email this data to his friend).

They all collaborate and bring their uniqueness or speciality to achieve a common goal.

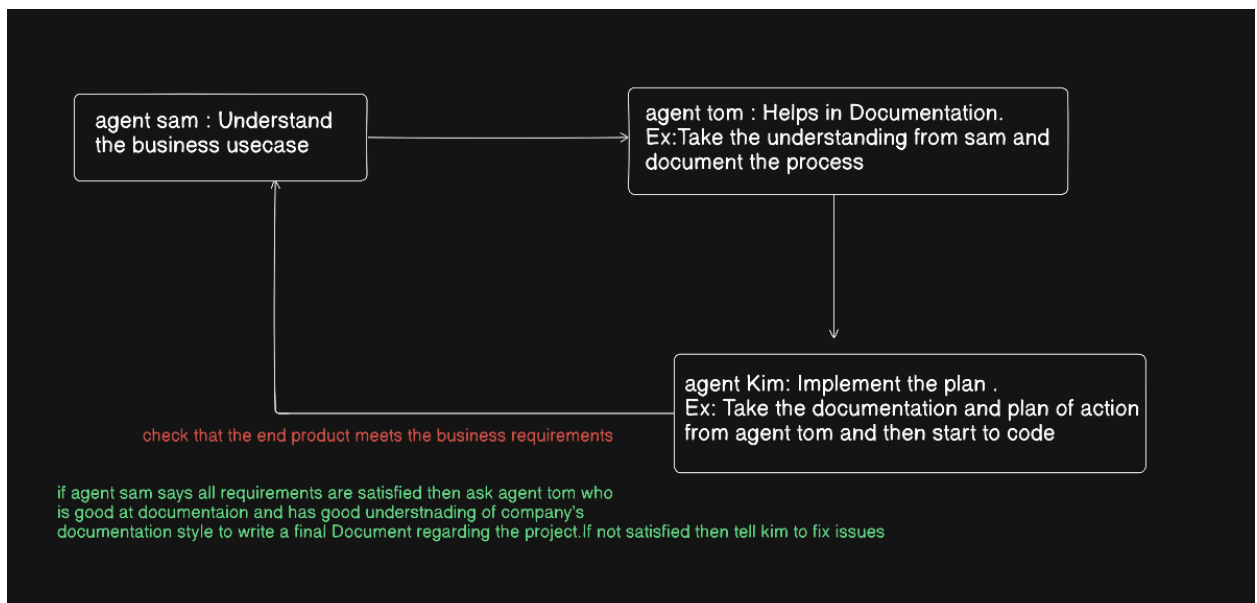
They may not always need to be group . It can also exist single when it is given the autonomy to do something like plan , decide , choose, etc.

One thing To highlight is that these agents become very powerful when proper tooling and utilization are done .

2. Workflow AI

Simple and clearcut understanding of workflow ai is how these ai agents are connected to each other as well as other entities. Example : agent 1 send the data to agent 2 . agent 2 modifies and then sends to agent 3.

Here is a workflow i created for demonstration.



3.AI copilots / assistants

AI copilots or assistants are the specialized agents that help us accompany a task. The most famous and widely used copilots are the github copilots and microsoft copilots. We can also say that they are like assistants which are sitting next to us and help us to achieve that task. One thing to keep in mind is that in AI copilots and assistants, humans are the most involved and the AI just assists and does not entirely make decisions.

4. Difference between a standalone AI model and AI integrated into a product workflow

A standalone AI model is a raw system that generates text based on a prompt or user input. Example: asking ChatGPT for cleaning dataset code.

Integrated AI is the AI model which is embedded into the systems which is able to perform tasks on a wider scope.

Example: integrated AI model can even run the cleaning dataset script on terminal.

Ex: Antigravity which has AI integrated with IDE.

Past experience you have with:

1. GenAI, LLMs, RAG, AI agents

- **Work Experience as SDE intern**

I worked on multiple POCs involving LLMs and GenAI. I experimented with models such as Qwen, Gemma, Meta Llama, and OpenAI models for tasks including automatic speech recognition, transcription analysis, log summarization, and insight generation.

1. So one of them was that I took the Selenium test run meta data and logs and integrated an AI model (local for security and privacy), gave those logs to the AI model and generated insights of test case runs.

2. I have also worked with audio-based models (Whisper, Qwen, NVIDIA models) for automatic speech recognition. Our internal CRM used voice chat so the model helped in transcription and analytics.

- **Personal Projects :**

1. Intel Industrial Project: Worked with Intel to develop a NCERT RAG for class 5 to 10 books. The main catch was that class 5 questions should not get class 10 answers.

Ex. Photosynthesis exists in both class 5 science and class 10 science book. A class 5 student won't understand class 10 answer. So I had to separate the vector database effectively and make the database separated.

2. Major Project is around **OIL AND GAS Industries with LAW and legislative constraints**. Creating a RAG which shall contain all the updated information about the laws and legislative (ex: Gov of India recently closed the access to this land due to crime issues.)

The vector database would be filled every 5 hours with trusted source of data and scrapped news ,using ETL pipelines and orchestration tools like Airflow.

3. AI agents : I have used frameworks like autogen [ag2] and Crewai for few projects:

https://github.com/shloktalhar25/Agents_with_autogen_and_crewAi

4. Good command on LLM traceability and observability using langfuse :

https://github.com/shloktalhar25/Langfuse_llm_observability

5 . created samuel code for claude code alternative in which 2 agent collaborate , interact and communicate together to give code as the user asks.

Ex: user asks for giving the basic code for fastapi application

AI model on cli : creates the directory and inserts the code into the folders without manual intervention .Github: <https://github.com/shloktalhar25/samuel-code>

2.Workflow automation

For workflow automation I have very good command . May it be Traditional SDE and testing automation: **Browser test automation using selenium** or even ETL pipelines workflows that automate data extraction, transformation, and loading . Also used Cloud services (azure , aws)

Created a MLOPS cicd pipeline for auto training model on real time iot data on azure infrastructure.

Worked with real time data streaming for ml , and ai

3.AI integration into a web platform or product

1.In my Previous internship I integrated small open source analytical model

(<https://huggingface.co/prem-research/prem-1B-SQL>) with internal CRM tool for their usecase.

2. Chatbot integration on web app

3 . Integrated Small language model with custom selenium testcase scripts for log analytics

4 . One very Interesting Integration i am currently working on (open soure project) is NLP models into DuckDB (engine) such that user can speak "I want to get record where in department is "Finance " and the model would generate the query and perform transformation on duckdb internally (duckdb is very strong with low latency).